

Question Paper

Exam Date & Time: 03-May-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL/MAY 2025

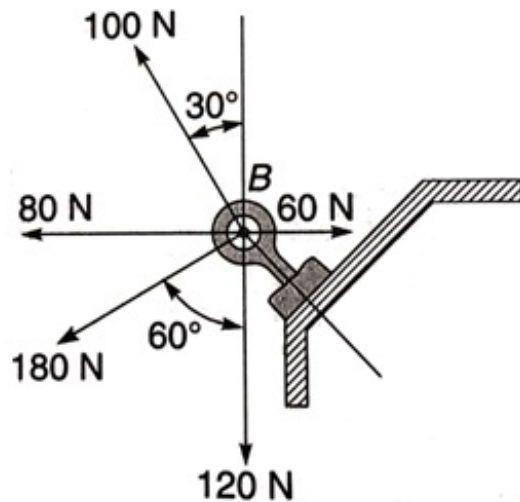
SUBJECT: CIE 1071/CIE_1071 - MECHANICS OF SOLIDS

Marks: 50

Duration: 180 mins.

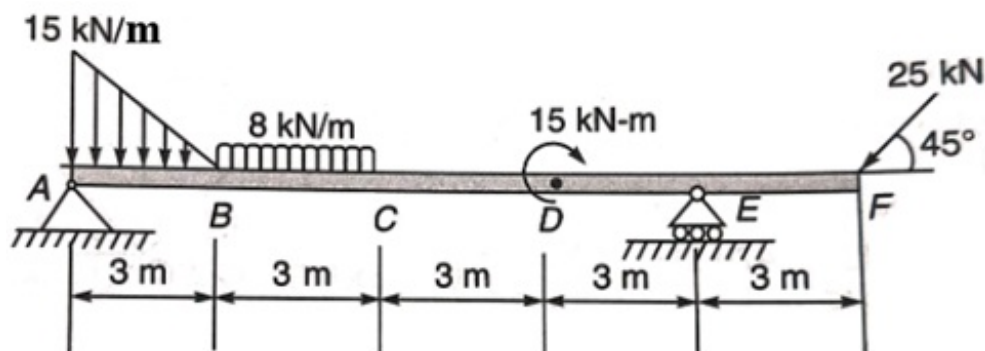
Answer all the questions.

- 1A) Five forces act on a bolt B as shown in the figure. Determine the resultant of the forces on the bolt (5)



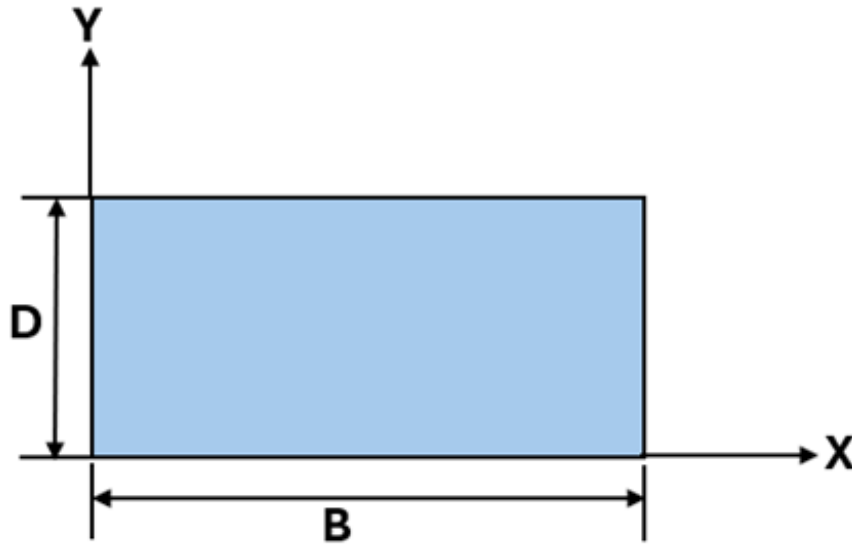
Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Determine the reaction at A and E for the beam subjected to loading as shown in figure. (5)



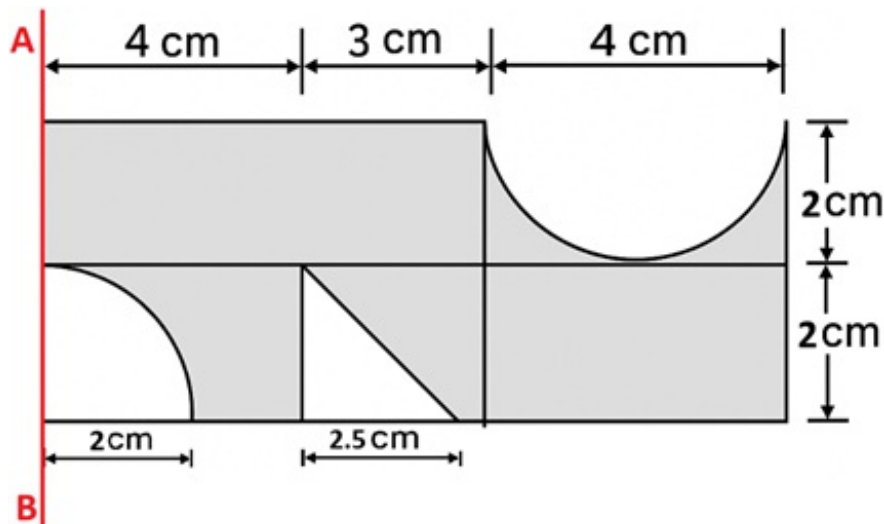
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2A) Show that the centroid of a rectangular area $B \times D$ is located at $(B/2, D/2)$ with respect to X and Y axis, using method of moments. (3)



Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2B) Determine the second moment of area of shaded area about given AB axis. All dimensions are in cm. (5)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) What is an axis of symmetry? Discuss its significance in relation to the centroid. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3A) A thin cylindrical vessel is 600 mm in diameter, 20 mm wall thickness and 0.8 m length. Determine the maximum pressure it can withstand, if the permissible tensile stress is 30 N/mm², permissible shear stress is 12 N/mm² and permissible change in diameter is 0.4 mm. Take $E = 200 \text{ GPa}$ and $\mu = 0.3$. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

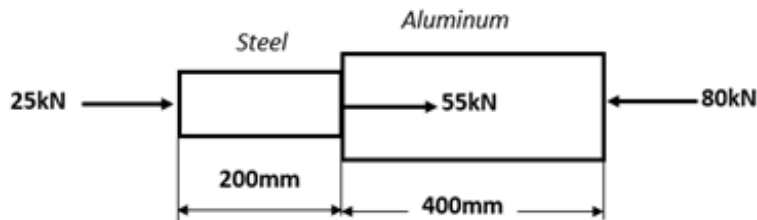
- 3B) Determine the relationship between modulus of elasticity (E), modulus of rigidity (G) and Poisson's ratio (μ), when a square element ABCD of side 'a' is subjected to pure shear ' τ '. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4A) A circular rod of 110 mm dia and 510 mm length is subjected to a tensile force of 2020 kN. Determine the modulus of rigidity, bulk modulus, and the volume change if the Poisson's ratio=0.3 and $E=2 \times 10^5 \text{ N/mm}^2$. (4)

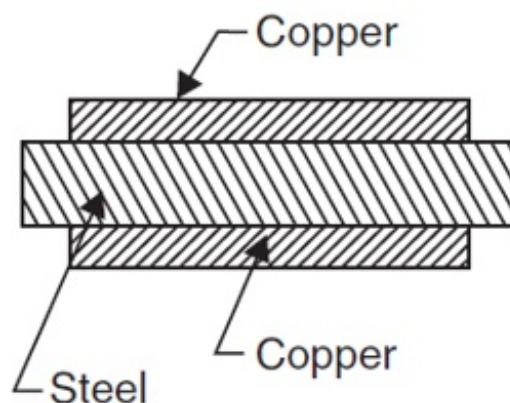
- 4B) Determine the relationship between volumetric strain and linear strain for a cube of side one unit subjected to three mutually perpendicular direct stresses. (3)

- 4C) A composite bar consists of a steel rod (20 mm diameter) rigidly fastened to an aluminium rod (60 mm diameter) as shown in figure. Axial loads are applied at the positions indicated. Determine the stress and strain in both rods. $E_s = 200 \text{ GPa}$ and $E_a = 70 \text{ GPa}$. (5)



- 5A) A steel rail is 12 m long and is laid at a temperature of 18°C . The maximum temperature expected is 40°C . (5)
- Estimate the minimum gap between two rails to be left so that the temperature stresses do not develop.
 - Calculate the temperature stresses developed in the rails if:
 - No expansion joint is provided.
 - If a 1.5 mm gap is provided for expansion.
- Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\alpha = 12 \times 10^{-6}/^\circ\text{C}$.

- 5B) A compound bar is made of a steel plate 50 mm wide and 10 mm thick to which copper plates of size 40 mm wide and 5 mm thick are connected rigidly on each side; the cross section of the assembly is as shown in Figure. The length of the compound bar at normal temperature is 1 m. If the temperature is raised by 80° , determine the stresses in each metal and the change in length. (5)
- Given $\alpha_s = 12 \times 10^{-6}/^\circ\text{C}$, $\alpha_c = 17 \times 10^{-6}/^\circ\text{C}$,
 $E_s = 2 \times 10^5 \text{ N/mm}^2$, $E_c = 1 \times 10^5 \text{ N/mm}^2$.



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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	20
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	9	40(80%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)